

Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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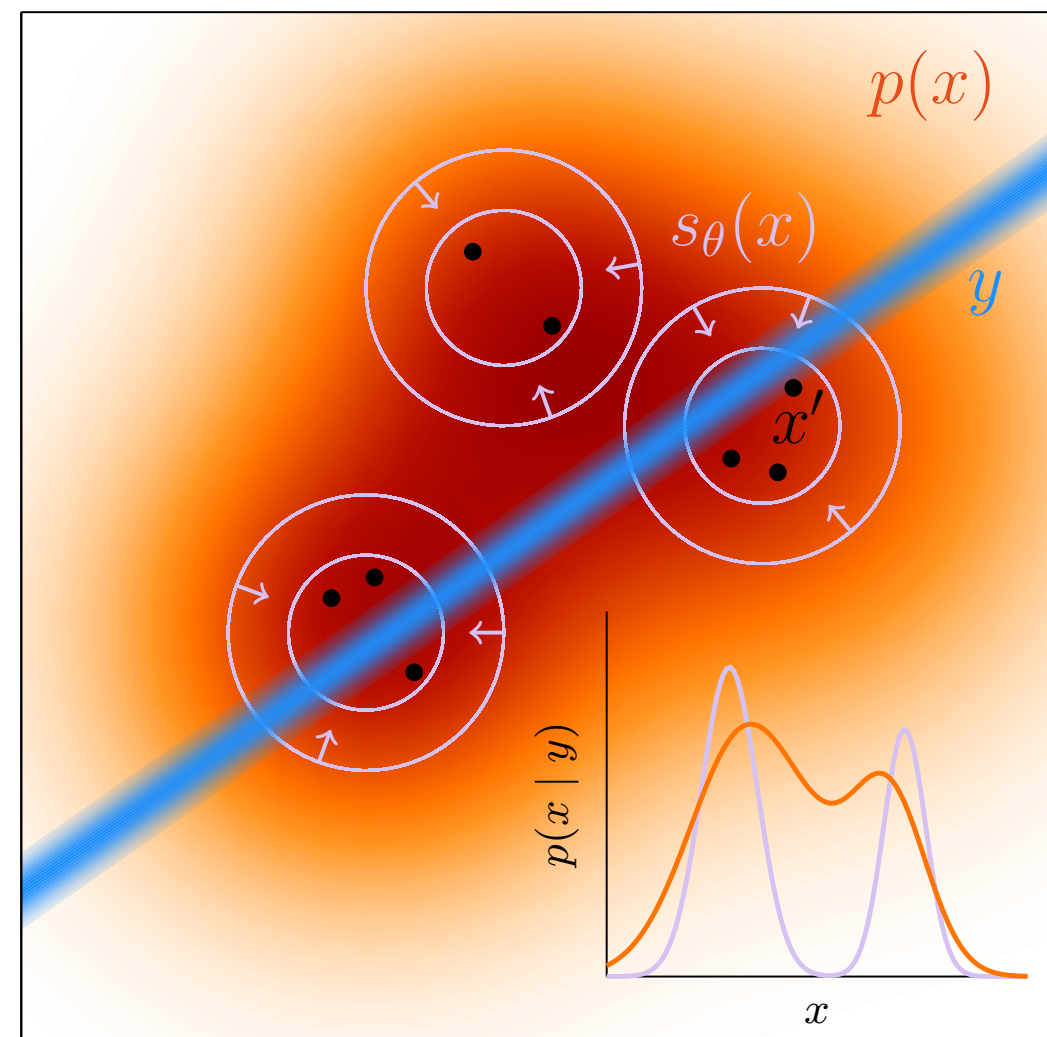
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The question

For inference, a trained diffusion model $s_\theta(x)$ becomes a prior over cosmological fields x . If the model has memorized its training data (black points), its score (pink) will not accurately reflect the true prior score, pushing inference to local clumps in data space. When confronted with observational data (blue), the inferred posterior distribution $p(x|y)$ will therefore be biased and overly confident.



- Novelty:** when do generated samples memorize training samples w.r.t. N_{train} ?
- Conditioning:** can we recover accurate posteriors on cosmological parameters, θ ?
- Fidelity:** do samples retain accurate statistical properties (1 & 2-pt functions)?

Data & models

- **CAMELS Multifield Dataset** (IllustrisTNG; Villaescusa-Navarro et al. 2021): 1000 simulations $\rightarrow$ 128 slices each $\rightarrow$ 128k HI (neutral hydrogen) maps of size 128×128 .
- Six cosmological parameters: $\theta = (\Omega_m, \sigma_8, A_{\text{SN1}}, A_{\text{AGN1}}, A_{\text{SN2}}, A_{\text{AGN2}})$.
- **Diffusion objective.** We corrupt a real map with noise and train a UNet to predict the velocity target at every noise level, with the cosmology θ as conditioning.

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, \epsilon, t} \|s_\theta(x_t, t, \theta) - \epsilon\|_2^2.$$

- **Sampling:** 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline.
- **Training-set sweep:** Train 3 models of increasing depth (3,4,5 layer UNet), each with 6 different training set sizes: $N_{\text{train}} = 2^6 \rightarrow 2^{15}$.

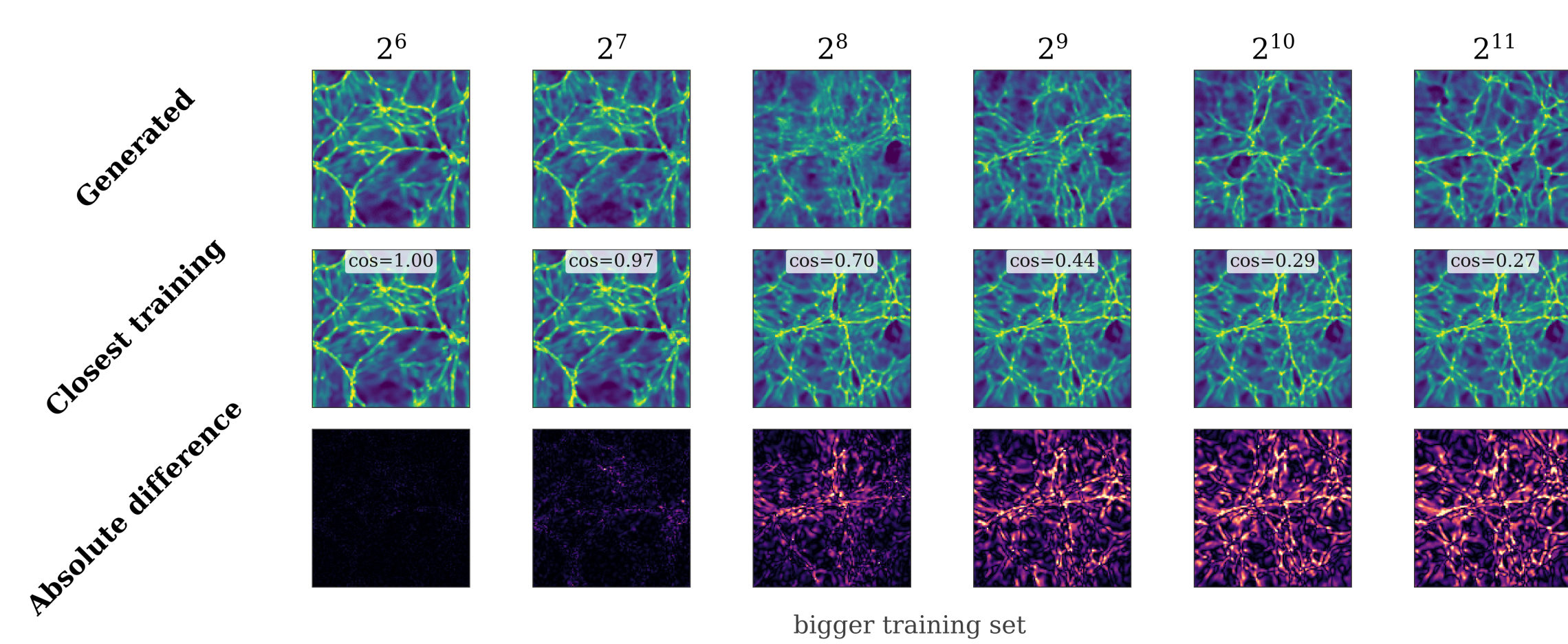
Detecting memorization

After training, we generate 100 novel 2D HI maps x_j , compare them to every map in the model's training set y_i , and keep the closest match:

$$s_j = \max_i \text{sim}(x_j, y_i)$$

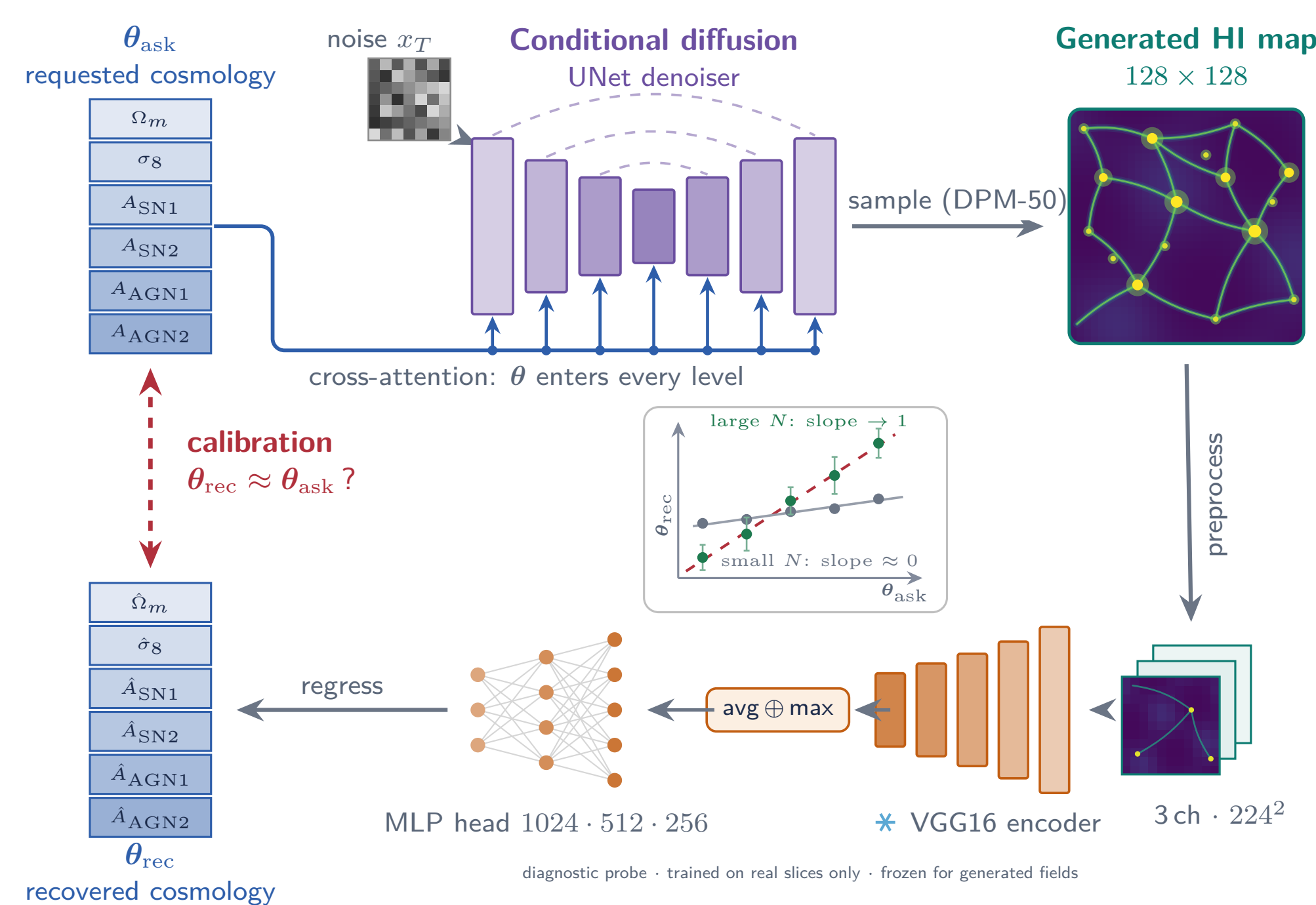
A high s_j means x_j is nearly a copy of one of its training maps. Separately, we train a PCA encoder to map 2D images into a vector space for classification. Here, sim is **cosine similarity** (larger = more alike) measured in either pixel or PCA latent space. For a fixed model size, the generated samples will transition from memorization to generalization at some N_{train} threshold.

UNet-64: generated samples and nearest training slices



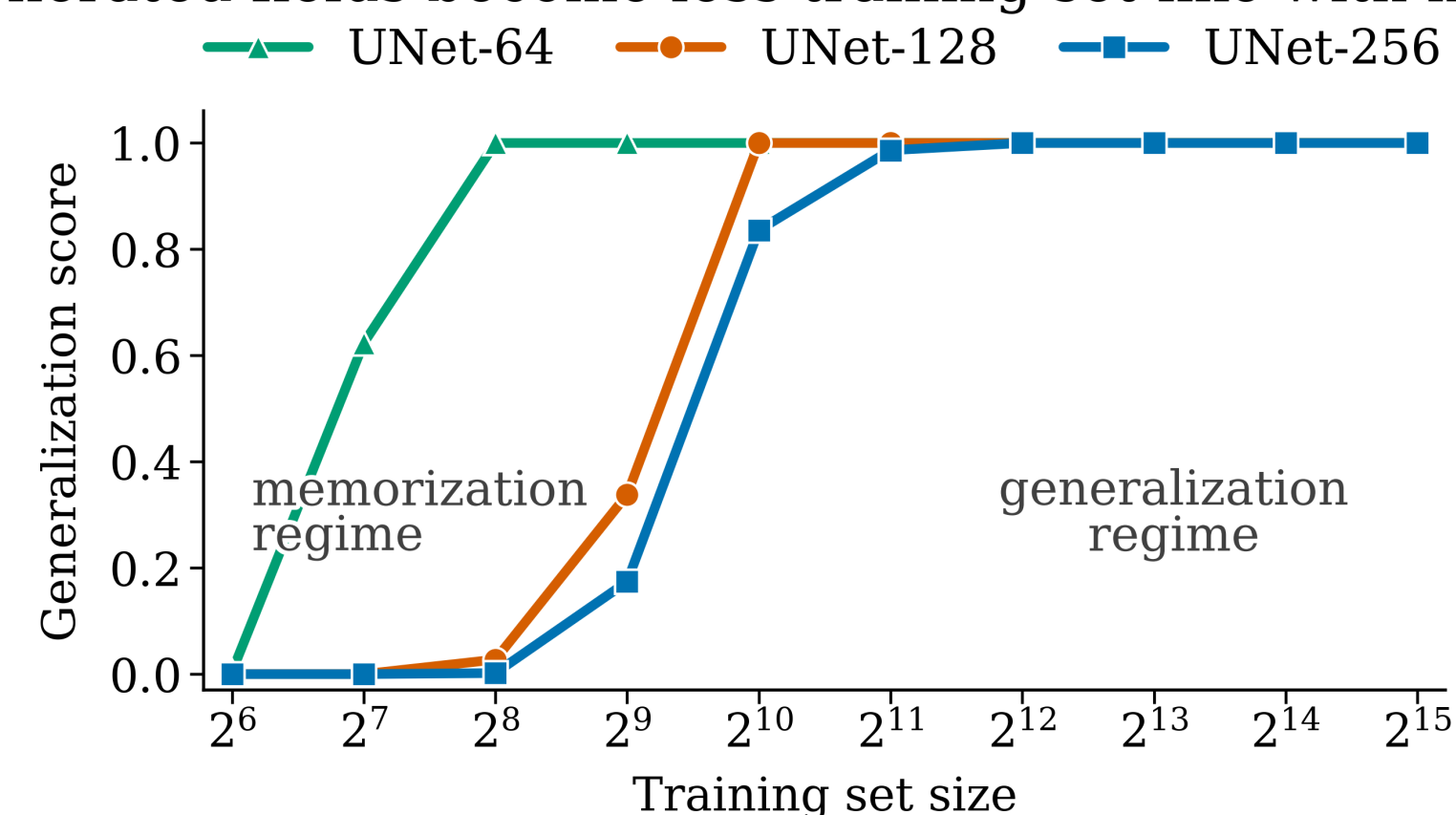
Inference: do we get the correct cosmology θ ?

We ask a conditional diffusion model to generate a sample with a given cosmology θ_{ask} . We pass those 2D samples to a classifier (VGG16) with an MLP head $\rightarrow \theta_{\text{rec}}$, trained on our full 128k dataset. Thus, we are able to compare the actual cosmology generated by the diffusion models θ_{rec} when conditioned on an input cosmology θ_{ask} , which we probe as a function of training set size and model capacity.



Memorization $\rightarrow$ generalization transition

Generated fields become less training-set-like with more data



For each case we generate 1000 samples: the generalization score is the fraction of generated maps that are *not* close to any training map,

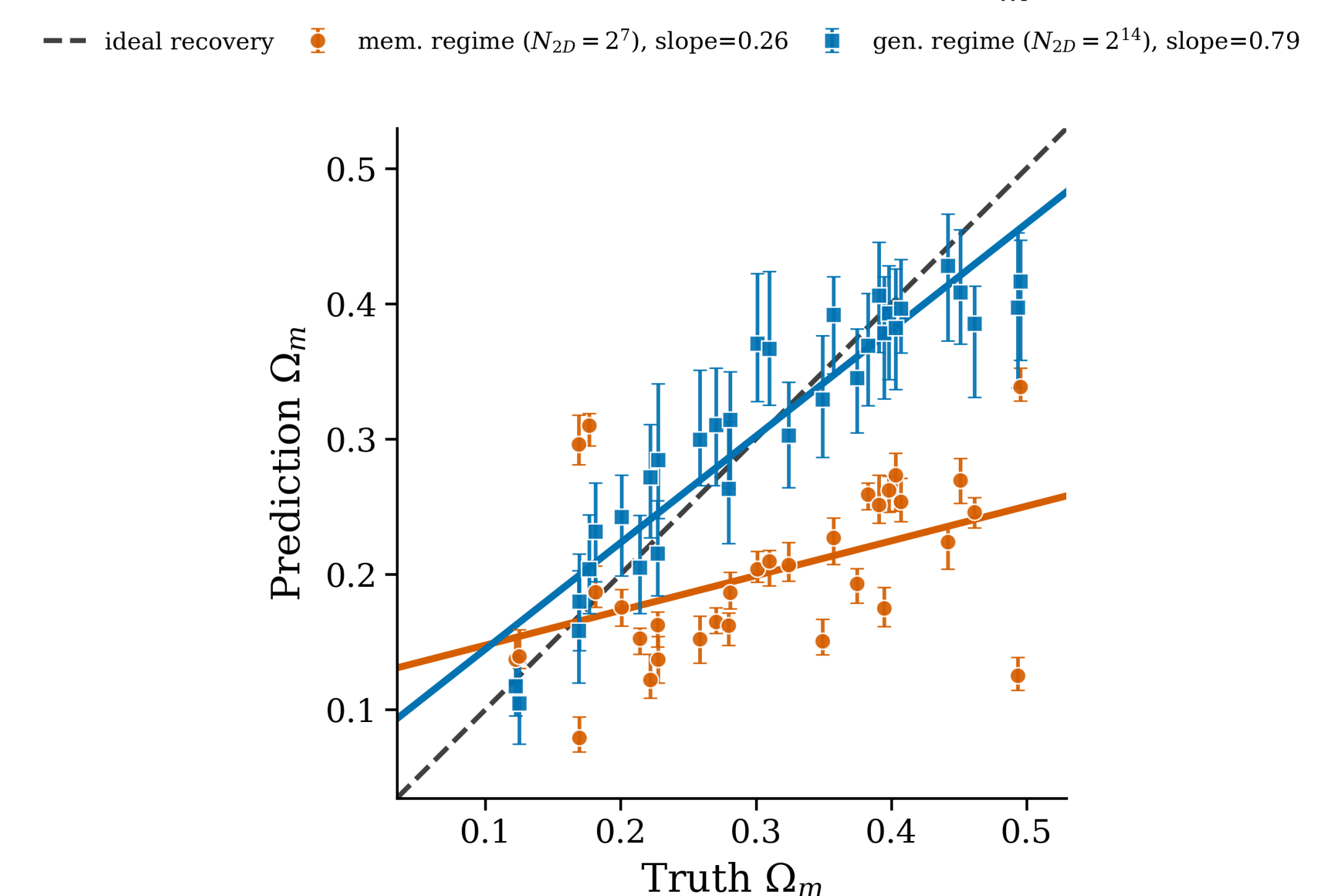
$$G = \Pr[s_j < \tau],$$

based on the cosine similarity score s_j , with τ (the “memorization gap”) set by the 95th percentile of train-vs-train closest-match similarity.

- $G \approx 0$: generated samples = training samples (**memorization**).
- $G \approx 1$: generated samples are unique to training samples.
- The transition happens at larger N_{train} for larger models (is this regime predictable and universal?) (Zhang et al. 2024; Kadkhodaie et al. 2024; Yoon et al. 2023). Currently exploring DiTs and other cosmo datasets.

Calibration results

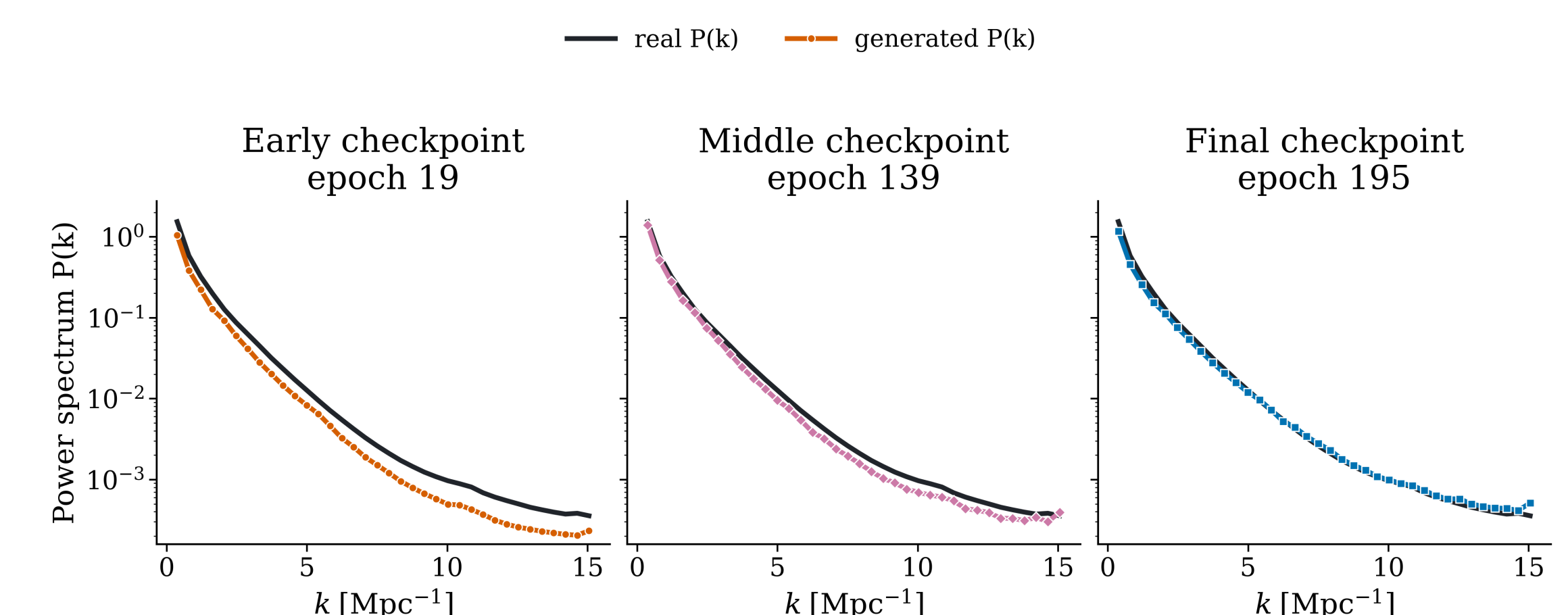
Generated HI fields recover Ω_m



Main result: we generate 1000 Ω_m -conditional samples from two models and predict the actual Ω_m in the maps: the errorbars show 16–84% scatter at fixed θ_{ask} . In the generalization regime (blue), the recovered Ω_m is within the true Ω_m 95% of the time. In the memorization regime (orange), the recovered parameters are more biased and overconfident. The VGG+MLP encoder has $R^2 = 0.91$ on real held-out Ω_m , so the difference is a true reflection of the diffusion model.

Statistical fidelity

Generated power spectra approach the real reference during training



Here we show that our diffusion models are indeed retaining accurate statistical information in the maps (via the 2-pt function aka power spectrum). During training, we show the first learns large-scale features and fills-in small-scale features later in training.

Takeaways

- The generalization transition appears in cosmological data.** The known memorization-to-generalization transition also occurs in CAMELS cosmological maps.
- It biases inference.** Small training sets yield models that memorize: inference looks precise but is biased toward the training values. Past the transition, recovered cosmology θ has better calibrated posteriors.
- Heuristic for practitioners.** Larger models generalize at bigger training set sizes: can we predict this transition across model & dataset types?